

Keeping Strategic Futures Observable in AI Strategy: Counterfactual Observability and Evidence Architecture under Radical Uncertainty

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Abstract

Artificial intelligence is not only being used strategically but as a recurring infrastructure to generate, compare, monitor, and refine alternatives in strategy. Research has shown that AI can broaden attention, speed feedback, improve strategic reasoning, support analogies, and enable some types of foresight. However, there are many other risks associated with the use of AI in strategy. For example, after one future enters execution, AI-enabled strategy systems have the capability to redistribute evidence toward the enacted path and away from serious non-enacted futures. I define this risk as the counterfactual observability gap the widening difference in the rate, richness, and review visibility of decision-relevant evidence generated for an enacted future compared to a serious shadow future after provisional commitment. The gap depletes the strategic futures reserve available at any point in time, defined as the stock of alternatives that remain legible, legitimate, and practically reversible under radical uncertainty. My central claim is that AI-enabled strategy can fail by recommending the wrong path ex ante, or by changing how evidence is collected, organized, and presented ex post so that rival paths become evidentially silent before uncertainty has materially receded. I develop a formalized theoretical model in which AI increases the velocity of evidence produced for the enacted path while survival of shadow futures depends upon proxy evidence, sponsorship, and retained capability for re-entry. I then specify counterfactual evidence architecture as a governance response: separable comparison dashboards, bounded probes, anomaly-triggered re-estimation, shadow sponsorship, reserve budgets, modular capability retention, and dual cadence. The article contributes to AI and strategy, real options, attention-based theory, path dependence, strategy-as-practice, and AI epistemic-regime research by shifting the dependent variable from option generation to option survival. It concludes with falsifiable hypotheses, a flagship randomized experiment, archival event-history tests, construct measures, public-case illustrations, and a managerial protocol for less self-erasing strategy systems.

Keywords: artificial intelligence; strategy process; radical uncertainty; strategic foresight; counterfactual observability; strategic futures reserve; evidence architecture; practical reversibility; real options; path dependence

Highlights

- AI can expand early strategic search while shortening later option survival.
- Chosen paths become more observable because action produces data; shadow futures require deliberately produced proxy evidence.
- Strategic futures reserve captures whether serious alternatives remain legible, legitimate, and practically reversible.
- Practical reversibility, not formal option existence alone, predicts whether a firm can pivot intelligently after late shocks.
- Counterfactual evidence architecture keeps at least one serious rival future learnable while execution proceeds on another.

1. Introduction: from option generation to option survival

Strategy managers continue to employ artificial intelligence as an additional tool to create and evaluate potential future scenarios, summarize markets, compare competitors, identify and assess risk areas, prioritize alternative strategic options, monitor implementation, and update dashboards. In essence, there is an underlying assumption that these tools will help strategy teams become more adaptive. While this optimism may be justified, it does not represent a complete picture of how artificial intelligence affects the strategy development process. The greatest impact of artificial intelligence may not occur during the generation of possible courses of action, but after a selected course of action is implemented into ongoing organizational processes.

When a preliminary decision regarding a future direction for the business is made, it sets off a series of events and actions that begin to produce evidence. This evidence includes progress toward specific objectives, customer responses to products or services, technical issues, signs of talent shortages or employee-performance constraints, operating expenses, technology usage data, product telemetry, capital funding requirements, model revisions, and management narratives about organizational activity and decision making. These signals eventually become part of the “data” that later artificial intelligence systems analyze, synthesize, predict, and display. On the other hand, a serious non-chosen alternative future will likely produce very little additional evidence over time unless the organization deliberately maintains some level of knowledge about what might have been. This may require shadow pilots, analogical benchmarks, external sensing, technical spikes, partner or customer probes, regulatory scanning, or revised assumptions about causality.

This issue matters because of the nature of strategy under conditions of radical uncertainty. Strategy development under significant uncertainty about the environment and available opportunities is not merely the task of selecting among known options at time zero. Instead, when many potentially viable future states cannot be explicitly enumerated, when the stability of probabilities used to estimate each option is questionable, when feedback is delayed or noisy, and when reversing previously made commitments is difficult, intelligent strategy requires a sufficient stock of viable alternatives that can permit revision as experience accumulates after the original decision is made (Knight, 1921; Rindova & Courtney, 2020; Townsend et al., 2025). Therefore, a company can still fail despite having originally committed to a viable future path.

Research in the developing field of AI and strategy has progressed significantly. Proprietary AI systems can increase attentional control during both strategy creation and implementation (Laamanen et al., 2025). LLMs can change mental representations under time pressure, although they do not appear to produce reliable improvements in strategic foresight (Kanis et al., 2026). Strategy simulations provide a basis for comparing AI performance across varying levels of complexity and delayed feedback (Allen & McDonald, 2026), and LLMs can retrieve very large sets of analogical information while still struggling to establish causal relationships (Sen et al., 2026). For agentic systems to facilitate strategy, they need architecture, decomposed elements, and relatively stable intermediate products (Camuffo et al., 2026). Formal work has also shown how AI agents can learn by using actual paths realized by firms while superior alternatives remain unexplored (Neshenko & Ryall, 2026).

However, these studies focus primarily on upstream issues surrounding how serious alternatives are generated before an initial AI-supported path begins to guide action. In contrast, this study focuses on a downstream issue: what occurs to serious alternatives once an initial AI-supported path begins to influence action? I theorize this issue as the counterfactual observability gap. The term “counterfactual” does not refer

to an ability of firms to observe an alternate world. Rather, it refers to a comparative evidence problem. Paths implemented by firms produce on-path evidence through action, whereas shadow futures require intentionally created proxy evidence to maintain their learnability. If no proxy streams are available, shadow futures tend to age, become more susceptible to political pressure, and become more difficult to re-enter.

The primary organizational state studied in this article is strategic futures reserve: a conceptual construct representing the stock of serious alternatives that exist within an organization at a particular moment in time and remain legible, legitimate, and practically reversible. Legibility means that a future remains represented within narratives, causal maps, models, scenarios, artifacts, and decision routines that actors can interpret and discuss. Legitimacy means that credible actors continue to sponsor the future without fear of being labeled disloyal, unrealistic, or outdated. Practical reversibility means that sufficient evidence, capabilities, interfaces, partners, budget, and trigger logic exist so that the firm may re-enter the previously abandoned future without rebuilding from scratch.

The article's central claim is narrow but significant: AI-enabled strategy systems may improve pre-commitment option generation while simultaneously reducing post-commitment option survivability when evidence generation, metricization, and review cadence become linked to the enacted path. This article does not contend that AI is inherently detrimental to strategy. Instead, it suggests that AI alters the evidential landscape of strategy. Under certain conditions, a system learning quickly about the selected path may learn less about what was not selected.

To mitigate this issue, I present counterfactual evidence architecture as a governance response. Its objective is not to ensure that all possible alternatives are preserved indefinitely or to celebrate indecisiveness. Rather, the purpose is disciplined pluralism under uncertainty: preserving enough diagnostic evidence for at least one serious shadow future while uncertainty remains material or until the shadow future fails a pre-defined viability test. Counterfactual evidence architecture includes separable comparison dashboards, bounded shadow probes, anomaly-triggered re-estimation, shadow sponsorship, reserve budgets, retained modular capabilities, and dual cadences.

This article contributes by shifting research in AI strategy from solely examining whether AI can create, evaluate, or implement strategic options to whether AI-enabled strategy systems provide adequate evidentiary environments for options to remain strategically viable. Thus, the unit of analysis expands beyond the strategic decision, algorithms, or formally defined options to include the future-by-review-cycle: whether a serious alternative receives sufficient diagnostic evidence, social sponsorship, and operational traction to endure successive cycles of AI-mediated evaluation.

The paper progresses as follows. Section 2 highlights differences from related literatures. Section 3 introduces the major constructs. Section 4 illustrates the formal intuition behind the distinction between formal and practical reversibility. Section 5 describes the AI-specific mechanisms that either widen or reduce counterfactual observability gaps. Section 6 presents hypotheses and competing predictions. Section 7 details counterfactual evidence architecture. Section 8 describes a multi-method empirical program. Section 9 uses publicly disclosed case examples as illustrative explanations for AI mechanisms. Section 10 explores boundary conditions and failure modes. Section 11 describes the contributions of this paper, and Section 12 concludes.

Figure 1: Core Mechanism of Counterfactual Observability

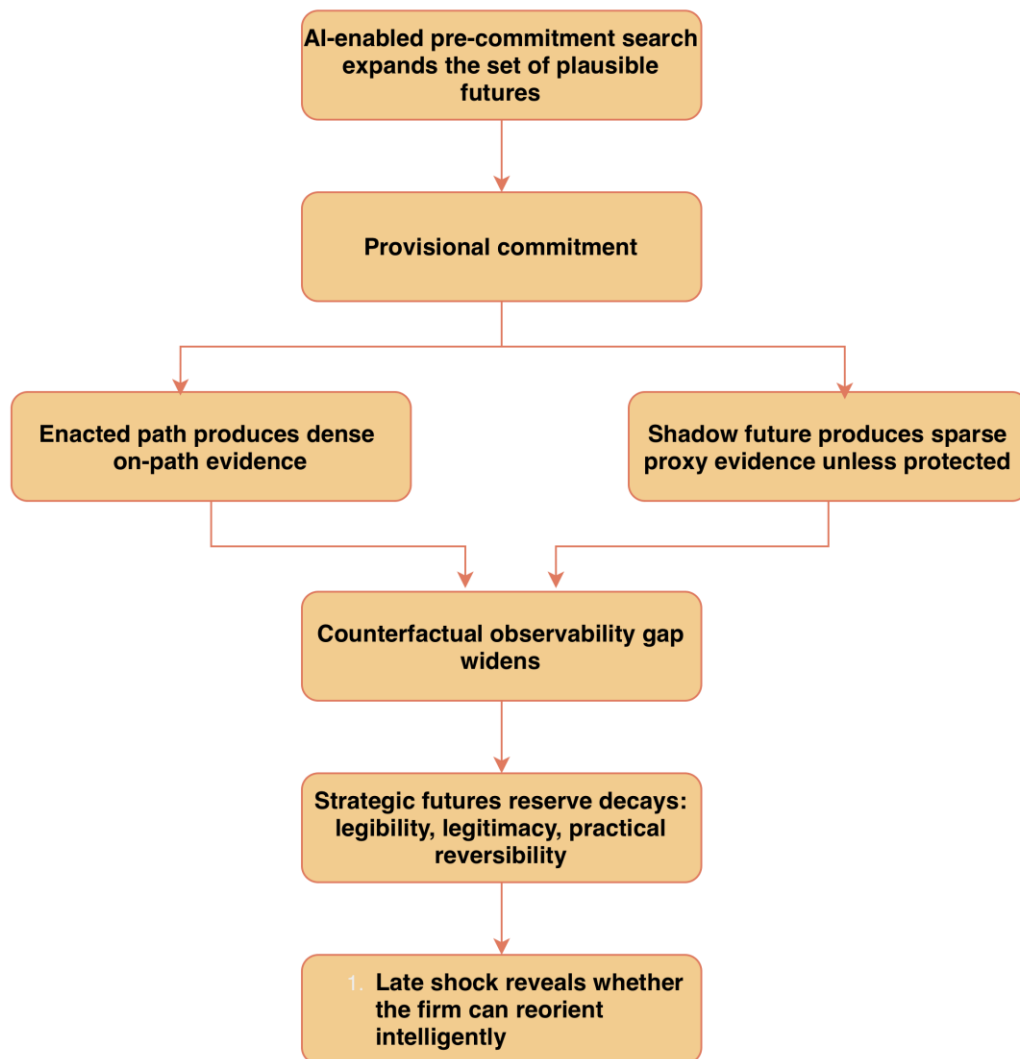


Figure 1. Core mechanism of counterfactual observability.

2. Why existing theories stop short

While I build upon, and do not reject, existing strategy and organization theories, I do not contend that prior research has ignored alternatives, flexibility, attention, scenarios, or lock-in. Instead, my contribution focuses on an underdeveloped temporal process: how evidence becomes distributed across enacted and non-enacted futures after strategic commitment. Existing literatures describe how alternatives are conceived, how attention is focused, how strategies are staged, how paths become locked in, and how AI alters authority and ways of knowing. However, they do not fully address how AI alters which futures continue to generate evidence after a provisional decision has begun to guide action.

The attention-based view explains how organizations direct their attention toward various issues, responses, and modes of communication (Ocasio, 1997; Ocasio & Joseph, 2018). Recent research has extended the attention-based view by showing how company-specific AI systems may expand the scope of attention, democratize the strategy-making process, and increase the speed of feedback cycles. However, I build on

these findings by asking a subsequent question: what occurs after attention converges? Equal breadth of attention at time zero does not guarantee equal preservation of evidence. Two organizations may develop the same set of alternatives at time zero; however, they may differ dramatically in whether shadow futures remain observable after dashboards, budget allocations, and review routines become associated with one enacted path.

Futures studies, scenario planning, strategic foresight, and related efforts help organizations generate possible futures and mitigate tunnel vision (Schoemaker, 1995; Csaszar & Laureiro-Martinez, 2018; Gavetti & Menon, 2016). Generative AI has altered the bottleneck, so the key question is not whether top managers can identify multiple potential futures, but whether any of those futures contain sufficient evidence to permit credible revisiting after commitments have been made. Thus, a comprehensive collection of future scenarios can coexist with a depleted futures reserve. An organization may recall that a particular future was previously identified while lacking the mechanisms necessary to revisit that future credibly.

Research on real options and strategic flexibility explains the benefits of delayed investment, flexible exit rights, staged investment, and partial commitment under uncertain conditions (McGrath & Nerkar, 2004). My theory adds an additional condition for the realization of option value: evidence. Although an organization may retain formal reversibility, meaning the legal or financial capacity to switch, it may experience diminishing practical reversibility. An organization may possess the ability to switch legally or financially; nevertheless, it may no longer have access to fresh evidence, credible sponsors, modular resources, or operational interfaces required to transition intelligently. Therefore, counterfactual observability identifies why formally reversible options can become empty shells before their expiration.

Research on path dependence and lock-in explains how self-reinforcing investments, complementary resources, increasing returns, narrowing action space, and eventually irreversible investment decisions occur (Sydow et al., 2009). Research on AI-induced lock-in demonstrates how AI agents learn from realized paths while superior alternatives are left unexplored. The counterfactual observability gap supplements this literature by identifying an earlier organizational process: alternatives can lose legitimacy and become less knowable before visible lock-in is completed. Reserve collapse is distinct from lock-in; it is often a mechanism through which lock-in becomes difficult to detect and reverse.

Research on open strategy and strategy-as-practice shows that participation, transparency, artifacts, visualizations, meetings, and situated routines shape strategic activity (Whittington, 1996; Hautz et al., 2017; Kaplan, 2011; Seidl et al., 2024). Participation alone cannot resolve the observability problem. A large number of participants can engage in an AI-based evaluation process that reduces all alternatives to one apparently authoritative recommendation. On the other hand, even a small number of participants can maintain significant pluralism if they preserve separate evidence streams and separate roles for reviewing counterfactuals. The critical issue is not openness in general terms, but whether strategy practices maintain evidence parity across reasonable alternative futures.

Finally, organization theory on AI shows that intelligent technologies create new epistemic regimes, predictive orientations, and artificial certainties (Anthony, 2018; Faraj et al., 2018; Leonardi & Leavell, 2026; Levina et al., 2025; Sergeeva et al., 2026). These studies are important because they illustrate why AI-generated representations acquire authority and why detailed forecasts can be mistaken for knowledge. My theory describes this idea as a strategy-process mechanism: AI does not simply alter what actors believe about the world; it also shapes which futures continue to produce evidence. Artificial certainty is particularly dangerous when attached to the only path that contains abundant amounts of data.

Table 1. Discriminating the counterfactual observability argument from adjacent theories.

Theory stream	What it explains	What remains underexplained	Discriminating prediction
Attention-based view	How attention is distributed across issues, answers, and channels.	Whether initially attended alternatives continue to receive diagnostic evidence after commitment.	Equal initial attention breadth can still yield faster reserve decay when enacted-path evidence dominates later reviews.
Scenario planning / foresight	How organizations imagine and compare plausible futures.	Why scenarios fade after budgets and dashboards attach to one path.	More scenarios at time zero will not preserve option survival unless shadow evidence is refreshed.
Real options	Value of staged investment, formal flexibility, and switching rights.	How evidence decay and sponsorship loss reduce practical exercise value.	Firms with equal formal reversibility diverge when only one preserves shadow-evidence parity.
Path dependence / lock-in	How self-reinforcing investments and complementarities narrow action.	How alternatives become less knowable before full economic lock-in is visible.	Observability gaps shorten future half-life even before sunk costs materially diverge.
Open strategy / practice	How participation, transparency, artifacts, and routines shape strategy.	Why inclusion may not prevent AI-mediated evidentiary closure.	Broad participation will not preserve alternatives when evaluation is monologic and metricized.
AI epistemic regimes	How AI changes authority, certainty, and ways of knowing.	How epistemic changes affect survival of rival futures over review cycles.	Artificial-certainty cues matter most when attached to the only path producing dense evidence.

3. Core theory: post-commitment evidence ecology

The theory is constructed using four load-bearing dimensions: AI strategizing embeddedness, the counterfactual observability gap, strategic futures reserve (SFR), and counterfactual evidence architecture. Other terms - future half-life, SFR ratio, shadow-evidence parity, practical reversibility, attrition, and foreclosure - are viewed as metrics, dynamics, or mechanisms rather than additional headline concepts. Reducing the number of headline concepts is important because the paper's contribution lies less in adding labels and more in identifying a single field-defining mechanism.

AI strategizing embeddedness (AISE) represents the degree to which AI is repeatedly utilized to assist in creating, assessing, authorizing, examining, and tracking strategic alternatives. Embeddedness is not merely having access to a chatbot or completing a one-time analysis. Instead, it describes the positioning of AI within repeated strategy microstructures: prompts, scorecards, simulations, decision copilots, agents, dashboards, implementation controls, review cycles, and model-updated planning routines. In this way, the theory has greatest application when AI becomes an integral component of the strategy system rather than an occasionally employed analytical tool.

The counterfactual observability gap (COG) is the time-varying disparity in decision-relevant evidence developed to support an enacted future versus a seriously contemplated shadow future following preliminary commitment. The basis for COG rests on a modest epistemological claim. Organizations cannot directly perceive unrealized worlds; however, they can create surrogate evidence about them through shadow pilot projects, simulation exercises, analogical comparisons, external sensing systems, technology spikes, partner probes, regulatory scans, customer interviews, or revised causal assumptions. The COG grows larger when enacted-path evidence develops faster than shadow-path evidence.

Strategic futures reserve (SFR) is the stock of serious alternatives that remain legible, legitimate, and practically reversible at time t . Serious alternatives are those that are plausible, consequential, causally distinct from the enacted pathway, and expensive to recreate later. Therefore, serious alternatives do not refer to all conceivable futures. A remembered future without a champion is ephemeral. A supported future without operational footholds is rhetorical. A recoverable future without evidence is opaque. SFR thus reflects whether a future remains organizationally alive rather than merely remembered.

Counterfactual evidence architecture (CEA) is the organizational infrastructure that maintains sufficient diagnostic evidence, sponsorship, and re-entry capacity to preserve the viability of at least one serious shadow future while the organization proceeds with disciplined commitment. CEA does not represent an instruction to hedge indefinitely. Rather, it defines a governance structure for periods of material uncertainty and active decision making.

Table 2. Core constructs, scope conditions, and observable indicators.

Construct	Definition	Scope condition	Observable indicators
AI strategizing embeddedness	Recurring use of AI in generation, evaluation, authorization, review, and monitoring of strategic alternatives.	AI is part of repeated strategy work, not a one-off aid.	Prompt logs; AI-generated artifacts; dashboard integration; agent workflows; scorecards; review scripts.
Counterfactual observability gap	Difference in rate, richness, and review visibility of evidence for enacted versus shadow futures.	At least two serious futures and a provisional commitment.	Evidence-density difference; dashboard coverage; update frequency; pilot records; quantified indicators; review airtime.
Strategic futures reserve	Serious alternatives that remain legible, legitimate, and practically reversible.	Most important under radical uncertainty and meaningful commitment costs.	Representational persistence; sponsor continuity; reserve budgets; trigger logic; modular capability retention.
Counterfactual evidence architecture	Routines and investments that preserve proxy evidence and re-entry capacity for a shadow future.	Useful when preservation costs are bounded and uncertainty remains unresolved.	Shadow probes; separate comparison dashboard; counterfactual review seat; anomaly triggers; reserve budgets.

4. Formal intuition: from formal options to practical reversibility

A formal core is necessary because the argument otherwise risks being absorbed into real options, attention-based theory, or path dependence. The central distinction is between formal reversibility and practical reversibility. Formal reversibility refers to the legal, financial, or contractual ability to switch away from a currently enacted path. Practical reversibility refers to whether the firm still possesses the evidence, legitimacy, and retained capability required to switch intelligently. The central claim is therefore not that AI removes formal options, but that AI-enabled strategy systems can erode the practical exercise value of those options even when the formal right to switch remains intact.

Consider a firm that has one enacted future, e , and at least one serious shadow future, s . Let $E_{j,t}$ denote the stock of diagnostic evidence available for future j at review cycle t , where j may refer either to the enacted future or to a shadow future. Evidence is not treated as raw data volume alone. A large volume of weak or non-comparable evidence should not count the same as smaller but highly diagnostic evidence. Therefore,

evidence can be represented as a weighted index of evidence volume, diagnostic quality, update frequency, quantification, review visibility, and causal comparability.

$$E_{j,t} = w_1x_{1,j,t} + w_2x_{2,j,t} + \dots + w_Mx_{M,j,t}$$

where $x_{m,j,t}$ is the normalized score of evidence component m for future j at review cycle t , and w_m is the weight attached to that component. This formulation clarifies that the theory concerns the diagnostic evidence environment surrounding a future, not merely the number of data points attached to it.

The counterfactual observability gap for shadow future s can then be represented as the log difference between enacted-path evidence and shadow-path evidence.

$$G_{s,t} = \ln(1 + E_{e,t}) - \ln(1 + E_{s,t})$$

The logarithmic form captures diminishing marginal informational value. The first few units of diagnostic evidence may substantially improve a future's observability, while additional evidence has declining marginal value. Because only an enacted-path advantage threatens the survival of the shadow future, the relevant depletion pressure is the positive part of the gap.

$$G_{pos} = \max(G, 0)$$

This distinction matters. A shadow future is not damaged simply because its evidence differs from the enacted future's evidence. It is damaged when the enacted future becomes substantially more visible, current, quantified, and reviewable than the shadow future.

Evidence evolves differently across enacted and shadow futures. The enacted path receives evidence because action itself produces observations. Shadow futures receive evidence only when the organization deliberately creates proxy evidence through shadow probes, external sensing, simulations, analogical benchmarks, technical spikes, partner tests, regulatory scans, or counterfactual dashboards. Evidence also depreciates because market conditions change, assumptions become stale, sponsors leave, and technical knowledge loses relevance. A compact representation is:

$$E_{e,t+1} = (1 - \delta_e)E_{e,t} + \lambda_{e(A_t, M_t, R_t)} + \varepsilon_{e,t}$$

$$E_{s,t+1} = (1 - \delta_s)E_{s,t} + \lambda_{s(A_t, B_{s,t}, P_{s,t}, D_{s,t})} + \varepsilon_{s,t}$$

where A is AI strategizing embeddedness, M is metricization intensity, R is review cadence, $B_{s,t}$ is the protected reserve budget for shadow future s , $P_{s,t}$ is the intensity of shadow probing, $D_{s,t}$ is dedicated counterfactual design, δ_j is the evidence obsolescence rate for future j , and $\varepsilon_{j,t}$ captures random disturbances affecting evidence production.

The enacted-path evidence function is assumed to increase with AI embeddedness, metricization, and review cadence.

$$\frac{\partial \lambda_e}{\partial A} > 0, \quad \frac{\partial \lambda_e}{\partial M} > 0, \quad \frac{\partial \lambda_e}{\partial R} > 0$$

The shadow-path evidence function increases with reserve budget, shadow probing, and counterfactual design.

$$\frac{\partial \lambda_s}{\partial B} > 0, \quad \frac{\partial \lambda_s}{\partial P} > 0, \quad \frac{\partial \lambda_s}{\partial D} > 0$$

AI has an asymmetric effect. When AI is primarily attached to execution monitoring, dashboards, and enacted-path review, it increases λ_e more than λ_s . However, when AI is deliberately embedded in counterfactual evidence architecture, it can also increase shadow-path evidence by supporting weak-signal detection, analogical search, simulation, anomaly detection, and shadow-dashboard updating.

$$D_{\{s,t\}} = 0 \Rightarrow \frac{\partial \lambda_s}{\partial A} \approx 0$$

$$D_{\{s,t\}} > 0 \Rightarrow \frac{\partial \lambda_s}{\partial A} > 0$$

Thus, AI is not inherently harmful or beneficial. Its effect depends on whether it is coupled only to the enacted path or also to the preservation of serious shadow futures.

The observability gap widens when the net growth of enacted-path evidence exceeds the net growth of shadow-path evidence. Using a first-order approximation, the change in the observability gap is:

$$\Delta G_{\{s,t\}} \approx \frac{\lambda_e(A_t, M_t, R_t) - \delta_{eE_{\{e,t\}}}}{1 + E_{\{e,t\}}} - \frac{\lambda_s(A_t, B_{\{s,t\}}, P_{\{s,t\}}, D_{\{s,t\}}) - \delta_{sE_{\{s,t\}}}}{1 + E_{\{s,t\}}}$$

The gap increases when:

$$\frac{\lambda_e(A_t, M_t, R_t) - \delta_{eE_{\{e,t\}}}}{1 + E_{\{e,t\}}} > \frac{\lambda_s(A_t, B_{\{s,t\}}, P_{\{s,t\}}, D_{\{s,t\}}) - \delta_{sE_{\{s,t\}}}}{1 + E_{\{s,t\}}}$$

This condition captures the central mechanism. AI-enabled strategy can increase the total amount of evidence available to the firm while decreasing the comparability of evidence across futures. In other words, the organization may learn faster about the future it has enacted while becoming less able to evaluate futures it did not enact.

Strategic futures reserve depends on whether serious shadow futures remain legible, legitimate, and practically reversible. Let $L_{s,t}$ denote the legibility of shadow future s , $Z_{s,t}$ its legitimacy, and $C_{s,t}$ its retained capability. Legibility means the future remains represented in current narratives, causal maps, models, scenarios, and review artifacts. Legitimacy means credible actors can continue to sponsor the future without being treated as disloyal, unrealistic, or outdated. Retained capability means the organization still has enough skills, interfaces, modular assets, partner access, documentation, and budgetary footholds to re-enter the future without rebuilding from zero.

The notation below uses romanized Greek terms such as rho, alpha, beta, gamma, and eta for Word compatibility. The three reserve dimensions evolve as follows:

$$L_{s,t+1} = \rho_L L_{s,t} - \alpha_L G_{pos,s,t} + \beta_L S_{p,s,t} + \gamma_L D_{s,t} + \eta_{L,t}$$

$$Z_{s,t+1} = \rho_Z Z_{s,t} - \alpha_Z G_{pos,s,t} + \beta_Z S_{p,s,t} + \gamma_Z D_{s,t} + \eta_{Z,t}$$

$$C_{s,t+1} = \rho_C C_{s,t} - \alpha_C G_{pos,s,t} + \beta_C B_{s,t} + \gamma_C D_{s,t} + \eta_{C,t}$$

where $Sp_{s,t}$ represents sponsorship support for the shadow future, $B_{s,t}$ represents protected reserve resources, $D_{s,t}$ represents counterfactual design, and $\eta_{L,t}$, $\eta_{Z,t}$, and $\eta_{C,t}$ are random disturbances. The persistence terms ρ_L , ρ_Z , and ρ_C capture the fact that legibility, legitimacy, and retained capability usually carry over from one review cycle to the next but may decay over time. The parameters α_L , α_Z , and α_C capture depletion caused by the observability gap, while the beta and gamma terms capture the protective effects of sponsorship, reserve resources, and counterfactual design.

Practical reversibility is not identical to formal reversibility. A firm may have the contractual right to switch but still lack fresh evidence, legitimacy, and retained capability. Let FR_s denote formal reversibility for shadow future s , normalized between 0 and 1. Let $Q_{s,t}$ denote a bounded transformation of shadow evidence.

$$Q_{s,t} = 1 - \exp(-kE_{s,t}), k > 0$$

This transformation means that additional shadow evidence improves practical reversibility, but with diminishing returns. Practical reversibility can then be represented as:

$$PR = FR \times Q^a \times Z^b \times C^c$$

where PR is practical reversibility for a given shadow future at a given review cycle, FR is formal reversibility, Q is bounded shadow evidence, Z is legitimacy, C is retained capability, and a , b , and c represent the relative importance of evidence, legitimacy, and retained capability.

A shadow future remains part of the strategic futures reserve only if it remains above minimum viability thresholds. Let $I_{s,t}$ indicate whether shadow future s remains alive at review cycle t .

$$I_{\{s,t\}} = 1[L_{\{s,t\}} \geq \tau_L, Z_{\{s,t\}} \geq \tau_Z, PR_{\{s,t\}} \geq \tau_{\{PR\}}]$$

where τ_L , τ_Z , and τ_{PR} are threshold values for legibility, legitimacy, and practical reversibility. The total strategic futures reserve at time t is then:

$$SFR_t = I_{1t} + I_{2t} + \dots + I_{Nt}$$

where N is the number of serious shadow futures identified at provisional commitment. The strategic futures reserve ratio is:

$$SFRRatio_t = \frac{SFR_t}{SFR_0}$$

This ratio captures the proportion of serious shadow futures that remain viable relative to the number of serious shadow futures available at provisional commitment. Future half-life can be defined as the number of review cycles until a shadow future falls below the reserve threshold.

$$HL_s = t_s^*$$

where t_s^* is the first review cycle in which $I_{s,t} = 0$. This makes the dependent variable more precise. The theory is not only about whether firms generate alternatives at time zero. It is about whether those alternatives remain alive across review cycles.

The post-shock implication is that adaptation should depend more strongly on practical reversibility than on formal option existence alone. Let $Adapt_{s,t+k}$ denote the quality or speed of adaptation toward shadow future s after a shock at time t . A compact empirical representation is:

$$Adapt_{\{s,t+k\}} = \beta_0 + \beta_1 PR_{\{s,t\}} + \beta_2 Shock_t + \beta_3 (PR_{\{s,t\}} \times Shock_t) + \Gamma Controls_t + u_{\{t+k\}}$$

where $Shock_t$ captures the magnitude or relevance of the adverse shock, $Controls_t$ includes sunk costs, switching costs, initial option attractiveness, environmental volatility, and formal switching rights, and u_{t+k} is the error term. The core prediction is:

$$\beta_3 > 0$$

This means that when a late shock makes the enacted path less attractive, firms with higher pre-shock practical reversibility should reorient faster and perform better than firms whose shadow futures have become evidentially stale, illegitimate, or operationally unrecoverable.

For a binary pivot outcome, the same implication can be expressed as a logistic model.

$$\Pr(Pivot_{\{s,t+k\}} = 1) = \frac{1}{1 + e^{\left\{ -\left[\beta_0 + \beta_1 PR_{\{s,t\}} + \beta_2 Shock_t + \beta_3 (PR_{\{s,t\}} \times Shock_t) + \Gamma Controls_t \right] \right\}}}$$

The formal intuition yields three implications. First, AI strategizing embeddedness can simultaneously increase enacted-path learning and reduce shadow-option survival when AI is coupled to execution metrics more strongly than to counterfactual evidence production. Second, counterfactual observability gaps deplete strategic futures reserve by reducing the legibility, legitimacy, and retained capability of serious shadow futures. Third, counterfactual evidence architecture preserves practical reversibility by increasing shadow-path evidence production, protecting sponsorship, slowing capability decay, and keeping at least one serious rival future above a viability threshold. This distinguishes the theory from real-options accounts in which formal switching rights dominate, from attention-based accounts in which initial attention breadth is central, and from path-dependence accounts in which sunk-cost accumulation is the primary mechanism.

5. AI mechanisms that widen or narrow the observability gap

5.1. Generative expansion before commitment

Both LLMs and agentic systems, including decision-support systems, reduce the costs associated with creating alternative scenarios, analogies, business models, risk narratives, implementation pathways, and critiques. These cost reductions enable broader exploration when human teams are limited by time, attention, or their own domain-specific categories. However, the creation of multiple options merely sets the stage for strategic thinking. The key issue is whether the created alternatives continue to exist as viable options once an initial option has been selected and begins guiding the actions of a team.

5.2. Metricizing compression during comparison

Many AI systems can summarize, rank, score, and display alternatives using similar criteria. In doing so, they may provide structure and discipline to decision making. However, this process can also force alternatives that were developed from different causal logics, time frames, or evidence bases into a single evaluative framework. Under conditions of radical uncertainty, the most easily measured possible future is not necessarily the best possible future. Measurability should not be mistaken for quality.

5.3. Path-coupled evidence accumulation after enactment

As soon as a particular course of action is implemented, it generates data about its implementation, which can then be used by subsequent AI systems to assess progress toward goals. Alternatives to what was eventually implemented generate very little additional evidence unless they are specifically probed. This difference in the production of new data is the basis for the counterfactual observability gap: taking action generates evidence, and that evidence may later appear to demonstrate strategic value.

5.4. Artificial certainty in representation

The output of AI tools often appears richly detailed, clear, precise, and objective. When confidence levels, simulations, forecast intervals, or polished rationales are attached to the course of action being taken, stakeholders may confuse high-quality representation with superior knowledge. In this sense, artificial certainty is not simply a cognitive bias. It is also a mode of presentation that supports the legitimacy of whichever course of action is currently supported by the most evidence and instrumentation.

5.5. Cadence compression in review routines

AI enables more frequent review processes. Frequent review can improve control over execution activities. However, under conditions of radical uncertainty, it can also accelerate evidentiary imbalance. An enacted path reviewed weekly with current metrics will generally appear stronger than an unimplemented alternative reviewed quarterly through memory and judgment. Therefore, review speed accelerates the gap between enacted paths and shadow futures unless organizations develop two cadences: rapid execution monitoring and slower strategic re-estimation.

5.6. Architecture monologism versus role-differentiated pluralism

An AI tool with a monologic architecture that provides one polished recommendation will tend to collapse alternatives relatively quickly. By contrast, an architecture that separates generation, criticism, analogical search, evidence auditing, integration, and counterfactual review can behave more like a structured pluralism engine than a convergence engine.

Together, these mechanisms suggest that AI embeddedness does not have a uniformly positive or negative effect on strategic adaptation. AI may expand the number of alternatives before commitment while shortening the survival of alternatives after commitment. The effect depends on whether AI is configured as a convergence engine that intensifies enacted-path evidence, or as a pluralism engine that preserves evidence across serious alternatives.

Table 3. AI-specific mechanisms and their expected effects on strategic futures reserve.

Mechanism	AI-specific source	Reserve dimension threatened	Observable signature
Generative expansion	Low-cost production of scenarios, analogies, variants, and critiques.	Initially increases breadth.	More named futures in early decks, prompt logs, and AI outputs.
Metricizing compression	Rankings, scores, summaries, and common dashboards.	Legibility of distinct causal logics.	Heterogeneous futures collapse into one score or common metric grammar.
Path-coupled evidence accumulation	Enacted-path data are fed into AI review loops.	Legitimacy and practical reversibility.	Chosen path receives more current metrics, pilots, and updates than shadow paths.
Artificial certainty	Detailed outputs make one future appear more knowable.	Legitimacy.	Precise forecasts, confidence displays, polished rationales, single-best recommendation.
Cadence compression	AI makes frequent review cheaper and faster.	All reserve dimensions.	Dense execution reviews with little shadow evidence refresh.
Architecture monologism	Single-output AI suppresses role-differentiated disagreement.	Legibility and legitimacy.	One authoritative answer replaces plural causal frames.

6. Hypotheses: option survival as the dependent variable

The theory makes discriminating predictions that differ from uniformly optimistic views of AI, classic real-options logic, standard path dependence, attention-based accounts, and scenario planning. The key dependent variable is not initial decision quality alone. It is the survival of serious shadow futures across review cycles. Three measures are central: the reserve ratio, future half-life, and shadow-evidence parity.

The reserve ratio is the number of serious shadow futures still above the reserve threshold divided by the number identified at provisional commitment. Future half-life is the number of review cycles until a shadow future falls below threshold on legibility, legitimacy, and practical reversibility. Shadow-evidence parity is the ratio of proxy evidence generated for a shadow future to on-path evidence generated for the enacted future, weighted by diagnostic quality and review visibility.

H1. Greater AI strategizing embeddedness is positively associated with the number of strategic alternatives generated before provisional commitment.

H2. After provisional commitment, greater AI strategizing embeddedness is positively associated with the counterfactual observability gap when metricization intensity and execution-linked review cadence are high.

H3. The counterfactual observability gap mediates the relationship between AI strategizing embeddedness and strategic futures reserve depletion.

H4. Higher counterfactual observability gaps shorten the future half-life of serious shadow futures after controlling for formal reversibility, sunk costs, and switching costs.

H5. Plural, role-differentiated AI architectures reduce reserve depletion relative to monologic AI support, holding base model quality constant.

H6. Faster execution-linked review cadence strengthens the negative effect of the counterfactual observability gap on strategic futures reserve.

H7. Artificial-certainty cues strengthen the relationship between enacted-path evidence and perceived strategic validity above and beyond objective performance indicators.

H8. Counterfactual evidence architecture weakens the relationship between the counterfactual observability gap and strategic futures reserve depletion.

H9. Under late adverse shocks, organizations with higher strategic futures reserve reorient faster and achieve higher post-shock performance than matched organizations with lower reserve.

H10. Reserve depletion is nonlinear: once shadow-evidence parity falls below a viability threshold, sponsor exit and re-entry capacity decay reinforce one another.

Table 4. Hypotheses and rival predictions.

Hypothesis set	Focal prediction	Rival explanation to test against
H1-H2	AI expands early generation but may widen observability gaps after commitment.	Uniformly positive AI accounts should predict better later reserve as well as better initial search.
H3-H4	The observability gap depletes reserve and shortens future half-life independent of formal reversibility.	Real-options logic would expect formal switching rights to preserve option value.
H4	Shadow futures can die before large sunk-cost divergence.	Classic path dependence would expect lock-in to follow stronger increasing returns or complementarities.
H5	Plural AI architecture preserves reserve more than monologic AI.	Model-quality accounts would predict no architecture effect after base capability is controlled.
H7	Artificial certainty amplifies perceived validity of evidence-rich paths.	Pure information accounts would predict perceived validity to track objective evidence quality only.

H8-H9	Evidence architecture improves post-shock adaptation by preserving practical reversibility.	Scenario planning alone would predict initial scenario breadth to be sufficient.
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7. Counterfactual evidence architecture

The evidence architecture of counterfactuals is the design solution to the problem of observability. It does not aim to create a sense of omniscience. In fact, it aims to do the opposite. It does not ask organizations to remain in a state of indecision or to keep all alternatives open indefinitely. Instead, its goal is to prevent the selected option from acquiring a monopoly over outcome evidence while material uncertainty remains unresolved. It converts a shadow future from a remembered possibility into a learnable and re-enterable alternative.

The first principle is evidence separability. An organization's execution evidence must be separated from evidence about the relative merits of different future options. Metrics that indicate whether the chosen option is being executed are not evidence that other options have weakened. Therefore, an organizational strategy dashboard must distinguish between operational metrics used to track how well the chosen option is being executed and comparative metrics used to assess whether rival futures remain viable.

The second principle is shadow probing. A shadow future should receive limited but deliberate evidence production. In other words, it should receive a small amount of diagnostic information. This evidence could come from customer probes, external sensing streams, simulations, analogical benchmarks, technical spikes, conversations with partners, or other bounded experiments. The key idea is not to replicate the execution of the enacted option. Rather, it is to provide enough diagnostic evidence so that the shadow future can survive as more than rhetoric.

The third principle is anomaly-triggered re-estimation. Before beginning execution on an option, the organization should identify specific events or conditions that would require it to compare current progress against the shadow future. These conditions might include missed thresholds, unexpected customer behavior, competitive actions, regulatory changes, model drift, capability bottlenecks, safety-related anomalies, or shifts in the organization's ecosystem. Specifying these events before execution begins reduces the political barrier to reconsidering an alternative option.

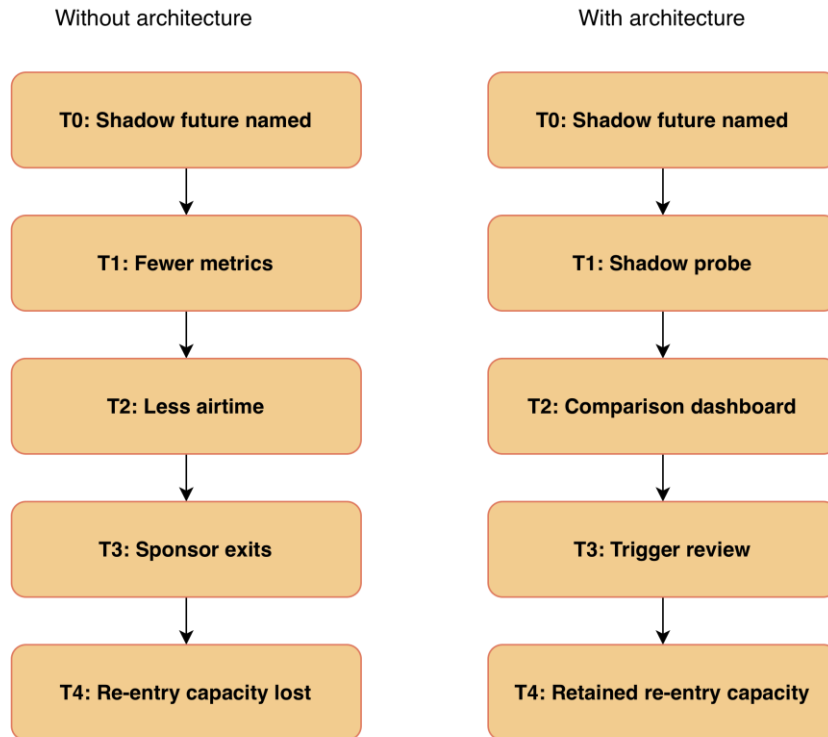
The fourth principle is bounded practical reversibility. Evidence alone is not enough. Even if evidence suggests that the chosen option is weakening, the organization must still be able to move toward an alternative. To make re-entry possible, organizations should reserve budgets, develop modular architectures, retain interfaces, establish contingent partnerships, document technologies and code, and preserve critical skills. These mechanisms prevent the shadow future from becoming impossible to reconstruct once the enacted path has absorbed most of the organization's attention and resources.

The fifth principle is shadow sponsorship. As part of each AI-mediated strategic review process, someone — a person, committee, or role-differentiated agent — should be assigned to interpret the results of shadow probes and explain what these results imply for the relative performance of competing options. Shadow sponsorship is not ritualized dissent. Rather, it is the sponsorship of a future option that would otherwise rapidly lose standing because fewer data speak on its behalf.

The sixth principle is dual cadence. Execution monitoring may need to occur frequently, especially when the enacted path is operationally complex. However, strategic re-estimation should not be collapsed into the same fast rhythm. If the enacted path is reviewed weekly through fresh metrics while the shadow future is

reviewed only through memory, the enacted path will almost always appear more legitimate. Dual cadence separates rapid execution monitoring from slower strategic reconsideration, allowing weak signals and shadow evidence to accumulate before alternatives are judged again.

Figure 2. Evidence architecture lengthens shadow-future half-life



Prediction: architecture preserves proxy evidence, sponsorship, and re-entry capacity above a viability threshold.

Figure 2. Counterfactual evidence architecture and shadow-future half-life.

Table 5. Managerial protocol for counterfactual evidence architecture.

Intervention	Purpose	Suggested metric	Protocol
Separate dashboards	Prevent execution KPIs from substituting for comparative evidence.	Shadow-evidence parity.	Maintain one execution dashboard and one comparison dashboard for serious shadow futures.
Shadow probes	Keep one rival future learnable.	Probe freshness and diagnosticity.	Run bounded probes each review cycle or every other cycle.
Anomaly triggers	Reopen comparison when reality deviates materially.	Trigger hit rate and response lag.	Pre-specify thresholds before execution begins.
Shadow sponsor	Preserve legitimate advocacy.	Sponsor continuity score.	Assign a formal role to interpret shadow evidence and maintain standing.
Reserve budget	Protect practical reversibility without duplicating the enacted path.	Reserve share of initiative spend.	Ring-fence modular resources and technical documentation for re-entry.
Dual cadence	Separate fast operational control from slower strategic reframing.	Review interval ratio.	Use frequent execution monitoring and periodic counterfactual re-estimation.

8. Empirical research program

A high-quality model based on this theory has to move beyond plausible concepts and anecdotal evidence. It requires a multi-method research program in which each study accomplishes something unique. At least one study should isolate the proposed mechanism. Another should demonstrate adaptive behavior over time under delayed disturbances. A third should validate the measurement system. A fourth should support organizational realism. Finally, at least one study should rule out alternative explanations.

The strongest experimental design uses the **future-by-review-cycle** as the primary unit of analysis. The main experiment should manipulate both the architecture of AI and the architecture of evidence. Participants are given a strategic fork involving one enacted path and two serious shadow futures. Participants select a preliminary path, receive sequential review-cycle evidence, and later encounter an adverse disturbance. The main experiment follows a 2x2 design: **AI architecture** is manipulated as monologic versus plural, and **evidence architecture** is manipulated as a single execution dashboard versus counterfactual evidence architecture. This design allows researchers to test whether AI compresses or pluralizes strategic thinking, and whether shadow futures survive when they continue to receive evidence after commitment.

In the **monologic-AI condition**, participants receive one polished recommendation, one aggregated rationale, and one integrated assessment per review cycle. In the **plural-AI condition**, participants receive role-differentiated outputs from a generator, critic, analogist, evidence auditor, and integrator, along with an explicit disagreement summary. These two conditions vary the architecture of AI reasoning while holding the underlying strategic problem constant.

The second manipulation concerns the architecture of evidence. In the **single-dashboard condition**, participants receive regular metrics only about the enacted path. The shadow futures remain formally available, but they receive no new proxy evidence during the review cycles. In the **counterfactual-evidence-architecture condition**, participants receive enacted-path metrics as well as bounded evidence about at least one serious shadow future. This may include a shadow dashboard, small probe results, updated external signals, anomaly triggers, and information about retained re-entry capacity. These two conditions vary whether shadow futures remain evidentially alive after provisional commitment.

The resulting four experimental conditions are: monologic AI with a single dashboard, monologic AI with counterfactual evidence architecture, plural AI with a single dashboard, and plural AI with counterfactual evidence architecture. The key prediction is that plural AI should reduce premature cognitive closure, while counterfactual evidence architecture should preserve shadow-future survival. The strongest adaptation should occur when plural AI architecture is combined with counterfactual evidence architecture, because this condition both pluralizes reasoning and preserves evidence for non-enacted futures.

The primary outcomes should include the counterfactual observability gap, strategic futures reserve, shadow-future half-life, shadow-evidence parity, pivot speed after the adverse disturbance, and post-shock performance. Secondary outcomes should include artificial certainty, perceived validity of the enacted path, confidence calibration, sponsor continuity, and explanation quality. This design allows researchers to test whether evidence architecture preserves option survival beyond the effect of AI architecture alone.

In addition to a laboratory-based design, a complementary archival design can study future survival rates across multiple strategy episodes. Researchers would first identify a focal enacted future and serious shadow futures from annual reports, investor presentations, earnings calls, strategy decks, product announcements, regulatory filings, patent records, job postings, alliances, and internal planning documents where available. A

reserve-collapse event occurs when a future loses representational presence, sponsor continuity, and operational re-entry capacity for two or more consecutive periods.

The most powerful field design would involve rolling out AI-assisted strategy tools or counterfactual evidence protocols across business units over time. Difference-in-differences and event-study designs would allow researchers to estimate whether AI-enabled monitoring increases evidence asymmetry and whether shadow-evidence protocols reduce reserve collapse. The strongest field evidence would combine dashboard histories, prompt logs, review artifacts, meeting records, budget allocations, staffing changes, and pilot portfolios.

Table 6. Flagship empirical design.

Design element	Publication-grade specification
Core design	2 x 2 randomized experiment: AI architecture (monologic vs role-differentiated plural) x evidence regime (single dashboard vs counterfactual evidence architecture).
Participants	480-560 screened managers, founders, product leaders, MBA/EMBA participants, or strategy professionals; executive replication recommended.
Task	Eight-round strategy simulation with one enacted path and two serious shadow futures under delayed and noisy feedback.
Shock	Late exogenous shock in rounds 6-7 that weakens the initially attractive path and improves a shadow path.
Primary outcomes	Reserve ratio, future half-life, counterfactual observability gap, practical reversibility, pivot speed, and post-shock payoff.
Secondary outcomes	Artificial-certainty index, perceived strategic validity, confidence calibration, sponsor continuity, explanation quality, and psychological ownership.
Models	Mixed-effects longitudinal models, discrete-time hazard models, mediation analysis, heterogeneous treatment effects, and robustness to alternative thresholds.
Transparency	Pre-register hypotheses, exclusions, power, manipulation checks, prompts, dashboard content, shock timing, and analysis scripts.

9. Measurement architecture

The construct system for this study must be validated rather than simply asserted. Future half-life, reserve ratio, shadow-evidence parity, and practical reversibility should be measured separately and then used to develop a predictive model of post-shock adaptation. The measures used to assess these factors must clearly differentiate them from generic attention, generic experimentation, indecision, scenario richness, formal flexibility, and sunk-cost accumulation.

A systematic approach to measuring these constructs begins with expert content validation. Strategy scholars and experienced senior strategists should judge whether each proposed indicator captures evidence, legibility, legitimacy, capability, formal reversibility, and/or generic attention. Interrater coding of artifacts should follow; subsequently, exploratory and confirmatory factor analyses may be conducted on survey measures. Ultimately, the predictive validity of the proposed indicators should be assessed through their ability to predict pivot speed, re-entry quality, and post-shock performance.

Finally, measurement cannot be tautological. Reserve should not be defined as later adaptation, and adaptation should not be inferred solely from reserve. The clean design for testing the theory is therefore temporal separation: first, code the level of reserve before the shock; then, measure both the speed at which the organization realigns itself and its overall post-shock performance.

Table 7. Measurement architecture for a top-tier empirical test.

Construct	Operational definition	Indicators	Validation step
Future half-life	Review cycles until a shadow future falls below viability threshold.	Agenda presence, refreshed assumptions, sponsor mentions, protected budget, retained capability.	Event-history validation; threshold sensitivity.
Reserve ratio	Share of serious futures still alive at time t relative to baseline.	Count of viable futures divided by baseline count, quality-weighted if needed.	Expert validation of serious-future classification.
Shadow-evidence parity	Relative diagnostic richness of shadow evidence versus enacted-path evidence.	Metrics, experiments, updates, simulations, external data, review airtime, causal-map revisions.	Convergent validity with expert evidence-quality ratings.
Practical reversibility	Ability to switch into a shadow future at acceptable cost and speed.	Talent, prototypes, modular assets, contracts, documentation, sponsor support, trigger logic.	Predictive validity for pivot speed and re-entry success.
Artificial certainty	Overconfidence induced by AI output form relative to calibrated uncertainty.	Unjustified precision, inevitability language, confidence display, willingness to terminate options.	Calibration tasks and manipulation checks.
Counterfactual evidence architecture	Design features intentionally preserving shadow evidence and re-entry capability.	Separate dashboard, probe budget, trigger register, review seat, reserve budget, dual cadence.	Nomological validity through moderation tests.

10. Public-case illustrations: mechanism signatures, not causal proof

Cases can help illustrate the mechanisms of the counterfactual observability gap, but they cannot bear the full weight of causal proof. Each case should therefore be used sparingly to demonstrate how one future can become increasingly instrumented, while another shadow future produces relatively less evidence. The theoretical argument is not that the enacted future was necessarily better or worse. Rather, the argument is that higher-density evidence is often mistaken for comparative superiority.

Tesla's autonomy strategy illustrates how an AI-native architecture commitment can generate dense on-path evidence through fleet data, model updates, evaluation infrastructure, hardware integration, and engineering routines. A competing autonomy architecture would need separate evidence production to remain learnable. This case is particularly helpful because it shows how a technical strategy can become increasingly observable through the same data infrastructure that enables it.

Meta illustrates both reserve depletion and reserve reconstruction. In 2021, Meta publicly repositioned itself toward creating the metaverse and articulated a metaverse-first corporate future. By 2024, Meta's disclosures highlighted accelerated capital expenditure to support its AI roadmap. This case is valuable because it demonstrates that strategic futures can change rank over time. Prior futures can regain legitimacy when external shocks, technological inflection points, or capital-market pressures change what evidence matters. Rebuilding reserves, however, is expensive.

Zillow Offers illustrates the risk of assuming that an instrumented enacted path has resolved uncertainty. Zillow scaled a data-intensive iBuying model and then announced in November 2021 that home-price forecast unpredictability and earnings volatility created too much risk to continue scaling further. Although the case also involves forecasting error, it is a vivid example of how a high-evidence enacted path can collapse when the evidence produced is not sufficiently comparative under volatile conditions.

GE's Predix strategy illustrates the industrial-transformation version of the same mechanism. In 2016, GE materials described Predix as a central operating platform for the Industrial Internet, supported by installed-base data and platform investments. This case demonstrates that observability is strongest where existing assets, customers, service contracts, and data streams already exist. That does not mean the broader platform future was impossible, but it changes which future becomes easier to evidence and govern.

Table 8. Public cases as observability-gap illustrations.

Case	Evidence-rich enacted future	Shadow future at risk	Theoretical use
Tesla autonomy	Vision/planning-centered AI autonomy architecture.	Multi-sensor or alternative deployment architecture.	Shows path-coupled evidence accumulation in an AI-native strategy.
Meta Platforms	Metaverse-first repositioning followed by AI infrastructure acceleration.	Alternative allocation across social, AI, AR/VR, and platform futures.	Shows reserve depletion and possible reserve reconstruction.
Zillow Offers	Algorithmic iBuying and digital transaction integration.	Asset-light marketplace future without balance-sheet exposure.	Shows evidence density can fail under volatile radical uncertainty.
GE Predix	Industrial Internet platform centered on Predix and digital industrial transformation.	Narrower asset-performance applications and partner-led ecosystem.	Shows observability concentrates where installed-base data already exist.

11. Boundary conditions, failure modes, and ethical limits

Theory design is consciously bounded by its limitations. It has greatest value when uncertainty is radical and commitments are high stakes; when feedback from experience is delayed or noisy; when AI systems are used recurrently throughout the strategy process; and when returning to a shadow future would be costly but not catastrophic. These conditions are less common in routine settings where state spaces remain relatively stable, probabilities remain reliable, experimentation feedback is rapid, switching between alternative paths is low-cost, and path dependence is weak.

In those routine settings, reducing the reserve of options may improve efficiency without undermining strategic goals.

This theory should not be interpreted as advocating for the continuous preservation of option space. Not all futures are worth protecting. Some alternatives are strategically weak, ethically unacceptable, legally impermissible, or too expensive to keep alive. The practitioner needs to protect only serious shadow futures: alternatives that are plausible, consequential, causally distinct, legitimately represented, legally permissible, and expensive to recreate later. Evidence architecture will be wasteful if used for trivial alternatives and dangerous if used to protect illegitimate ones.

Counterfactual evidence architecture can also fail. It may become symbolic if the counterfactual review seat has no authority to act on its findings. It may become performative if shadow probes are disconnected from actual decision making. It may become expensive if too many futures are preserved. It may become politically charged if managers use shadow futures as a way to delay action. The governance challenge is therefore disciplined pluralism: preserving enough serious alternatives to prevent premature foreclosure while maintaining commitment discipline.

The theory is also not antithetical to artificial intelligence. In certain settings, AI may help generate shadow evidence, monitor external analogies, detect weak signals, and trigger re-estimation. The issue is not AI use

itself, but asymmetric AI integration into the learning process. AI becomes dangerous when it intensifies knowledge about the chosen path while leaving rival futures inference-based, obsolete, and socially unacceptable.

Ethically, evidence architecture should not be used to protect futures that shift responsibility, violate safety laws, evade legal accountability, or hide harms under the language of option space. A future deserves reserve protection only if it is both strategically serious and ethically admissible. The architecture should therefore include an exclusion test: any alternative that does not pass legal, safety, rights, or legitimacy thresholds should not be protected as a shadow future.

Table 9. Alternative explanations and preemption strategy.

Reviewer objection	Why it is dangerous	Design preemption
This is just real options.	Reserve may look like renamed flexibility.	Hold formal reversibility constant while varying shadow-evidence parity.
This is just path dependence.	Lock-in literature is close.	Show reserve depletion before sunk costs and complementarities diverge materially.
This is just attention narrowing.	ABV could absorb the argument.	Equalize initial attention breadth and vary post-commitment evidence architecture.
This is not AI-specific.	Dashboards and planning systems also produce observability effects.	Include human-only and non-AI digital baselines; test AI x cadence x architecture interactions.
Cases are cherry-picked.	Public cases cannot prove causality.	Treat cases as illustrations and use preregistered experiments or archival panels for tests.
Reserve is indecision.	Preserving options may look like dithering.	Separate reserve from delay by measuring post-shock response quality and speed.
Measures are tautological.	Reserve could be defined by its outcomes.	Validate evidence, legitimacy, and capability indicators independently and predict later adaptation.
Managerial implications are generic.	Governance advice often becomes vague.	Tie each implication to a manipulated design feature and measured mechanism.

12. Discussion and contributions

This study contributes to the field of artificial intelligence and strategy by changing the dependent variable from decision quality and strategic capability to option survival. Most current research in this area focuses on what AI can generate, predict, evaluate, compare, or automate. These inquiries are important, but insufficient. In situations characterized by radical uncertainty, the primary strategic concern is not only that AI may make poor decisions. Rather, it is that AI-enabled strategy systems can alter the way evidence is organized, such that one path becomes increasingly validated while serious alternative paths become progressively more difficult to see.

This study contributes to strategy process and strategy-as-practice literature by treating AI as a recurring microstructure rather than as an episodic tool. Future depletion is not generated by a single recommendation. It arises from repeated associations among prompts, dashboards, review cycles, scorecards, budgeting processes, and implementation controls around a chosen path. Therefore, the theory describes how strategy systems become self-erasing: they provide evidence for the future they enact while forgetting how to discover information about futures they did not enact.

This study contributes to the literatures on strategic flexibility and real options by providing an evidentiary basis for option execution. Formal switching rights alone are not sufficient to confer practical reversibility. The ability to exercise an option intelligently also depends on whether shadow futures continue to develop new evidence, possess legitimate sponsors, and retain modular capabilities. Option value therefore depends

not only on formal rights or switching costs, but also on whether the organization continues to have sufficient knowledge, belief, and capacity to switch intelligently.

This study contributes to the literatures on path dependence and lock-in by describing a pre-lock-in mechanism. Rival futures can become less knowable before all switching costs have accumulated. By the time the organization realizes that the enacted path is weak, the rival future may no longer be representable, sponsored, or recoverable. Counterfactual observability therefore explains why organizations can become locked in not only by investments, but also by evidentiary asymmetries.

Finally, this study contributes to organization theory on AI by translating artificial certainty and epistemic-regime concepts into a specific strategy-process mechanism. AI-generated representations of futures do not merely influence beliefs; they influence which futures produce evidence and which require surrogate evidence to remain observable. Counterfactual evidence architectures therefore serve as a form of epistemic governance. They regulate the authority of AI outputs by preserving comparative uncertainty while uncertainty remains unresolved.

For managers, the implication is clear. The relevant issue is not who has developed the most capable AI model. Rather, it is who has created the least self-erasing strategy process. Organizations that establish counterfactual evidence architecture should recognize anomalous patterns earlier, preserve re-entry options more credibly, and recover from premature convergence more quickly than competitors. Because developing such a capability requires organizations to act intentionally while continuing to fund evidence against themselves, it is likely to be rare.

13. Conclusion

The problem facing strategy is not only about better analyzing what is happening. It is also about keeping track of which futures are still able to produce evidence as events unfold. Under radical uncertainty, this problem is central because organizations cannot know in advance which path will deserve dominance. Thus, the key issue is not only whether AI improves the ability to make choices, but whether AI-enabled strategy systems preserve enough comparable evidence for organizations to intelligently revisit and update their plans.

The counterfactual observability gap describes the fundamental asymmetry that emerges when enacted paths become increasingly visible while serious shadow futures do not. Strategic futures reserve describes the organizational state at risk. Practical reversibility describes the condition under which a formally available option remains usable or becomes hollow. Counterfactual evidence architecture describes the design response. Together, these terms explain how AI can enable future abundance at the front end of a strategy process while simultaneously eliminating viable options downstream.

Therefore, the most pernicious form of strategy is not the one that has too little imagination. Rather, it is the one that stops creating evidence for serious alternatives once one alternative appears to be emerging as dominant. The strategic potential of AI under radical uncertainty does not lie in determining which alternative will win first. Rather, it lies in developing AI-enabled strategy systems that allow alternative futures to remain legible, legitimate, practically reversible, and evidentially alive long enough to be properly assessed and revised.

References

- Allen, R. T., & McDonald, R. M. (2026). How well can AI do strategy? Empirical benchmarking using strategy simulations. *Strategy Science*, 11(1), 93-117. <https://doi.org/10.1287/stsc.2025.0444>
- Anthony, C. (2018). To question or accept? How status differences influence responses to new epistemic technologies in knowledge work. *Academy of Management Review*, 43(4), 661-679. <https://doi.org/10.5465/amr.2016.0334>
- Anthony, C., Bechky, B. A., & Fayard, A.-L. (2023). Collaborating with AI: Taking a system view to explore the future of work. *Organization Science*, 34(5), 1672-1694. <https://doi.org/10.1287/orsc.2022.1651>
- Boussioux, L., Lane, J. N., Zhang, M., Jacimovic, V., & Lakhani, K. R. (2024). The crowdless future? Generative AI and creative problem-solving. *Organization Science*, 35(5), 1589-1607. <https://doi.org/10.1287/orsc.2023.18430>
- Camuffo, A., Gambardella, A., Kazemi, S., & Pandey, A. (2026). Beyond black boxes: Designing and testing agentic AI systems for strategy. *Strategy Science*, 11(1), 137-156. <https://doi.org/10.1287/stsc.2025.0432>
- Chatterji, A., Csaszar, F. A., Evans, J., Felin, T., Hullman, J., Lakhani, K. R., Sako, M., & Zenger, T. (2026). Can AI do strategy? A dialogue and debate. *Strategy Science*, 11(1), 16-30. <https://doi.org/10.1287/stsc.2026.ed.v11.n1>
- Csaszar, F. A. (2018). What makes a decision strategic? Strategic representations. *Strategy Science*, 3(4), 606-619. <https://doi.org/10.1287/stsc.2018.0067>
- Csaszar, F. A., Hinrichs, N., & Heshmati, M. (2024). External representations in strategic decision-making: Understanding strategy's reliance on visuals. *Strategic Management Journal*, 45(11), 2191-2226. <https://doi.org/10.1002/smj.3613>
- Csaszar, F. A., Ketkar, H., & Kim, H. (2024). Artificial intelligence and strategic decision-making: Evidence from entrepreneurs and investors. *Strategy Science*, 9(4), 322-345. <https://doi.org/10.1287/stsc.2024.0190>
- Csaszar, F. A., & Laureiro-Martinez, D. (2018). Individual and organizational antecedents of strategic foresight: A representational approach. *Strategy Science*, 3(3), 513-532. <https://doi.org/10.1287/stsc.2017.0041>
- Csaszar, F. A., Lee, G., Zemsky, P., & Zenger, T. (2026). Can AI do strategy? *Strategy Science*, 11(1), 1-15. <https://doi.org/10.1287/stsc.2026.intro.v11.n1>
- Daft, R. L., & Weick, K. E. (1984). Toward a model of organizations as interpretation systems. *Academy of Management Review*, 9(2), 284-295. <https://doi.org/10.5465/amr.1984.4277657>
- Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*, 10(28), eadn5290. <https://doi.org/10.1126/sciadv.adn5290>
- Faraj, S., Pachidi, S., & Sayegh, K. (2018). Working and organizing in the age of the learning algorithm. *Information and Organization*, 28(1), 62-70. <https://doi.org/10.1016/j.infoandorg.2018.02.005>
- Gavetti, G., & Menon, A. (2016). Evolution cum agency: Toward a model of strategic foresight. *Strategy Science*, 1(3), 207-233. <https://doi.org/10.1287/stsc.2016.0018>
- General Electric. (2016). GE 2016 integrated report. https://www.ge.com/sites/default/files/GE_AR16_ISR.pdf
- Gerlach, J. P., & Lange, D. (2026). Fading memories: The role of machine learning in organizational knowledge depreciation. *Academy of Management Review*. <https://doi.org/10.5465/amr.2024.0408>
- Hautz, J., Seidl, D., & Whittington, R. (2017). Open strategy: Dimensions, dilemmas, dynamics. *Long Range Planning*, 50(3), 298-309. <https://doi.org/10.1016/j.lrp.2016.12.001>
- Kanis, T., Mann, J. E., & Stumpf-Wollersheim, J. (2026). AI-augmented strategic decision-making under time constraints: An experimental study on mental representations and strategic foresight. *Strategy Science*, 11(1), 75-92. <https://doi.org/10.1287/stsc.2025.0442>
- Kaplan, S. (2011). Strategy and PowerPoint: An inquiry into the epistemic culture and machinery of strategy making. *Organization Science*, 22(2), 320-346. <https://doi.org/10.1287/orsc.1100.0531>
- Kaplan, S., & Orlikowski, W. J. (2013). Temporal work in strategy making. *Organization Science*, 24(4), 965-995. <https://doi.org/10.1287/orsc.1120.0792>
- Knight, F. H. (1921). Risk, uncertainty and profit. Houghton Mifflin.

- Laamanen, T., Weiser, A.-K., von Krogh, G., & Ocasio, W. (2025). Artificial intelligence in adaptive strategy creation and implementation: Toward enhanced attentional control in strategy processes. *Long Range Planning*, 58(4), 102561. <https://doi.org/10.1016/j.lrp.2025.102561>
- Leonardi, P. M., & Leavell, V. (2026). Knowing enough to be dangerous: The problem of artificial certainty for expert authority when using AI for decision making and planning. *Organization Science*, 37(2), 516-543. <https://doi.org/10.1287/orsc.2023.18224>
- Levina, N., Gkeredakis, E., & Fayard, A.-L. (2025). Predictive data science as an epistemic stance: Benefits, risks, and opportunities for knowledge pursuit in organizations. *Strategic Organization*. <https://doi.org/10.1177/14761270251410676>
- McGrath, R. G., & Nerkar, A. (2004). Real options reasoning and a new look at the R&D investment strategies of pharmaceutical firms. *Strategic Management Journal*, 25(1), 1-21. <https://doi.org/10.1002/smj.358>
- Meta Platforms. (2021). Founder's letter, 2021. <https://about.fb.com/news/2021/10/founders-letter/>
- Meta Platforms. (2024). Meta reports first quarter 2024 results. <https://investor.atmeta.com/investor-news/press-release-details/2024/Meta-Reports-First-Quarter-2024-Results/default.aspx>
- Milliken, F. J. (1987). Three types of perceived uncertainty about the environment. *Academy of Management Review*, 12(1), 133-143. <https://doi.org/10.5465/amr.1987.4306502>
- Neshenko, N., & Ryall, M. D. (2026). When artificial intelligence does strategy: Learning, good times, lock-in, and human-driven strategic renewal. *Strategy Science*. <https://doi.org/10.1287/stsc.2025.0448>
- Ocasio, W. (1997). Towards an attention-based view of the firm. *Strategic Management Journal*, 18(S1), 187-206. [https://doi.org/10.1002/\(SICI\)1097-0266\(199707\)18:1+<187::AID-SMJ936>3.0.CO;2-K](https://doi.org/10.1002/(SICI)1097-0266(199707)18:1+<187::AID-SMJ936>3.0.CO;2-K)
- Ocasio, W., & Joseph, J. (2018). The attention-based view of great strategies. *Strategy Science*, 3(1), 289-294. <https://doi.org/10.1287/stsc.2018.0050>
- Raisch, S., & Krakowski, S. (2021). Artificial intelligence and management: The automation-augmentation paradox. *Academy of Management Review*, 46(1), 192-210. <https://doi.org/10.5465/amr.2018.0072>
- Rindova, V., & Courtney, H. (2020). To shape or adapt: Knowledge problems, epistemologies, and strategic postures under Knightian uncertainty. *Academy of Management Review*, 45(4), 787-807. <https://doi.org/10.5465/amr.2018.0291>
- Schoemaker, P. J. H. (1995). Scenario planning: A tool for strategic thinking. *Sloan Management Review*, 36(2), 25-40.
- Seidl, D., Ma, S., & Splitter, V. (2024). What makes activities strategic: Toward a new framework for strategy-as-practice research. *Strategic Management Journal*, 45(12), 2395-2419. <https://doi.org/10.1002/smj.3668>
- Sen, P., Workiewicz, M., & Puranam, P. (2026). Can LLMs aid analogical reasoning for strategic decisions? A comparative study. *Strategy Science*, 11(1), 118-136. <https://doi.org/10.1287/stsc.2025.0426>
- Sergeeva, A. V., Leonardi, P. M., & Faraj, S. (2026). Beyond the tool view of AI: Intelligent technologies and the emergence of new epistemic regimes. *Strategic Organization*. <https://doi.org/10.1177/14761270261436479>
- Sydow, J., Schreyogg, G., & Koch, J. (2009). Organizational path dependence: Opening the black box. *Academy of Management Review*, 34(4), 689-709. <https://doi.org/10.5465/amr.34.4.zok689>
- Tesla. (n.d.). AI & robotics. <https://www.tesla.com/AI>
- Townsend, D. M., Hunt, R. A., Rady, J., Manocha, P., & Jin, J.-H. (2025). Are the futures computable? Knightian uncertainty and artificial intelligence. *Academy of Management Review*, 50(2), 415-440. <https://doi.org/10.5465/amr.2022.0237>
- Weiser, A.-K., & von Krogh, G. (2023). Artificial intelligence and radical uncertainty. *European Management Review*, 20(4), 711-717. <https://doi.org/10.1111/emre.12630>
- Whittington, R. (1996). Strategy as practice. *Long Range Planning*, 29(5), 731-735. [https://doi.org/10.1016/0024-6301\(96\)00068-4](https://doi.org/10.1016/0024-6301(96)00068-4)
- Zillow Group. (2021). Zillow Group reports third-quarter 2021 financial results and shares plan to wind down Zillow Offers operations. <https://investors.zillowgroup.com/investors/news-and-events/news/news-details/2021/Zillow-Group-Reports-Third-Quarter-2021-Financial-Results--Shares-Plan-to-Wind-Down-Zillow-Offers-Operations/default.aspx>

Appendix

Appendix A. Construct measures and validation plan

A.1. Strategic futures reserve scale

Each serious shadow future can be rated after every review cycle. Legibility items: the shadow future remained represented in current strategy artifacts; the causal logic of the shadow future remained clear; managers could still explain how this future would create value. Legitimacy items: a credible actor remained willing to sponsor the future; invoking the future was treated as strategic rather than disloyal; the future received meaningful airtime in review. Practical reversibility items: resources existed to restart exploration; modular capabilities or partnerships were preserved; trigger logic for re-entry was explicit; managers could identify the first three actions required to re-enter the future.

A.2. Counterfactual observability gap index

The objective index should combine evidence volume, update frequency, quantification, dashboard coverage, pilot activity, causal-assumption refresh, and review airtime. For each review cycle, code chosen-path evidence and shadow-path evidence separately, standardize each component, and compute the difference. A perception scale should ask whether the chosen path received more current, quantified, credible, and decision-relevant evidence than the shadow future.

A.3. Artificial-certainty index

Items should capture whether AI outputs made the enacted path appear more precise, inevitable, objective, and validated than warranted by the evidence. The index should be discriminated from perceived model competence. A model can be competent while still producing artificial certainty when uncertainty is under-displayed.

A.4. Validation

The construct system should be validated through expert content review, exploratory and confirmatory factor analysis, interrater reliability for archival coding, discriminant validity from strategic flexibility and indecision, and predictive validity for post-shock adaptation. The most important validation test is whether shadow-evidence parity and practical reversibility predict post-shock reorientation after controlling for formal reversibility, sunk costs, scenario breadth, and initial option attractiveness.

Appendix B. Experimental protocol sketch

Participants receive a realistic strategic dossier describing a firm facing three plausible futures: an integrated proprietary path, a modular ecosystem path, and a partnership path. All three are initially serious, but they differ in time horizon, evidentiary base, and reversibility. Participants select a provisional path after receiving either monologic or role-differentiated AI support.

Over eight review cycles, participants allocate attention, budget, and sponsorship. In the single-dashboard condition, the enacted path receives rich metrics while shadow futures receive little new information. In the

counterfactual evidence architecture condition, a shadow dashboard, probe budget, anomaly triggers, and review seat preserve proxy evidence for one rival future. In rounds 6-7, a shock changes the payoff structure. The strongest evidence for the theory would show that evidence architecture preserves reserve before the shock and improves adaptation after the shock.

Primary outcomes are reserve ratio, future half-life, counterfactual observability gap, shadow-evidence parity, practical reversibility, pivot speed, and post-shock performance. Secondary outcomes include artificial certainty, confidence calibration, sponsor continuity, and explanation quality. The analysis should pre-register mediation tests and threshold robustness checks.

Appendix C. Archival coding dictionary

Appendix Table C1. Future-by-review-cycle coding dictionary.

Variable	Coding rule	Example sources
Future identification	A future is serious if it is named, causally distinct, sponsored, and resource-relevant.	Annual reports, investor days, strategy decks, earnings calls, board materials.
Legibility	Count and quality of artifacts representing the future.	Narrative sections, causal maps, product roadmaps, model outputs, scenario files.
Legitimacy	Presence and status of sponsors, executive mentions, board or committee attention.	Transcripts, org charts, governance charters, announcements, meeting minutes.
Practical reversibility	Evidence of retained capabilities, budgets, partnerships, modular assets, triggers, and documentation.	Budgets, pilot portfolios, patents, contracts, technical repositories, staffing plans.
Observability gap	Difference between chosen-path evidence and shadow-path evidence.	Dashboard metrics, telemetry, pilot reports, quantified filings, review artifacts.
Reserve collapse	Future falls below threshold on legibility, legitimacy, and practical reversibility for two consecutive periods.	Longitudinal coding across quarters or review cycles.